

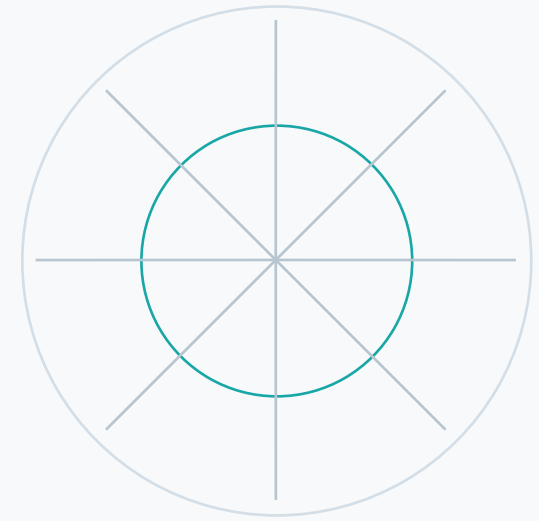
Cyclistic 2019 Bike-Share Analysis

How do annual members and casual riders use Cyclistic bikes differently?

Full-year 2019 scope • 3.8M trips • MySQL + Excel

Case-study objective

Understand distinct usage patterns and translate them into testable membership-conversion actions.



Business task & analytical approach

The case study asks for evidence that can support membership-focused marketing decisions.

BUSINESS TASK

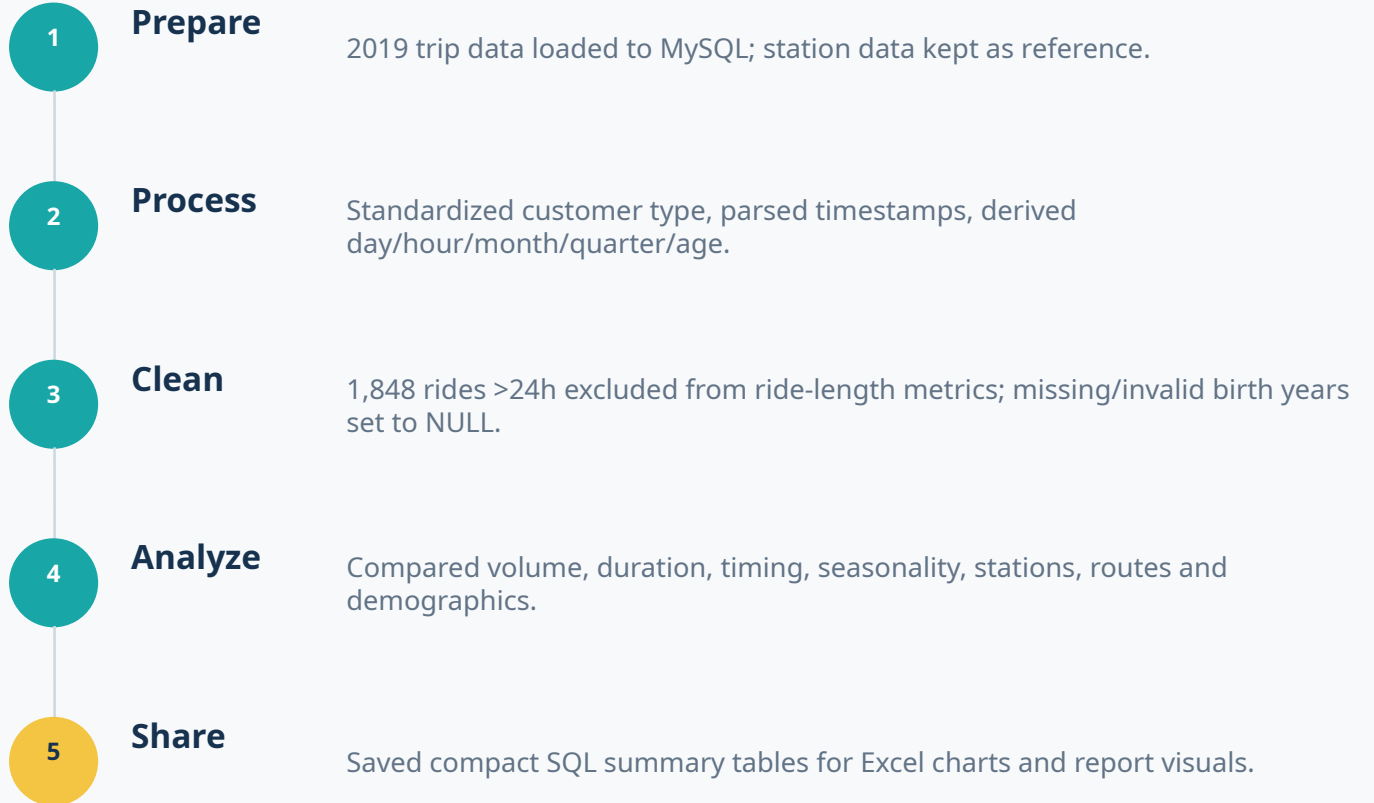
How do Members and Casual riders use Cyclistic bikes differently?

Primary stakeholder

Marketing leadership

Desired outcome

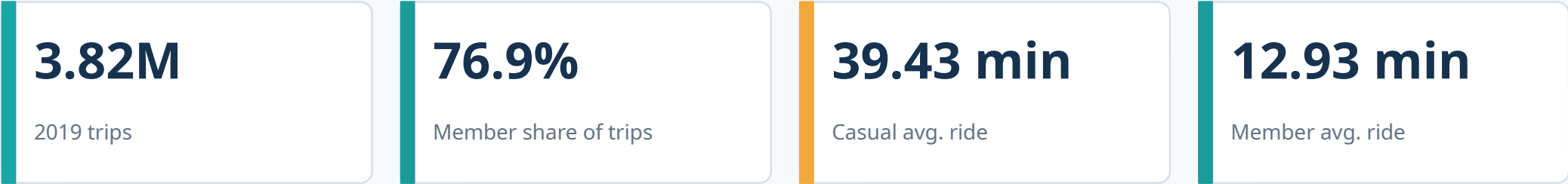
Identify behavior differences → design conversion tests



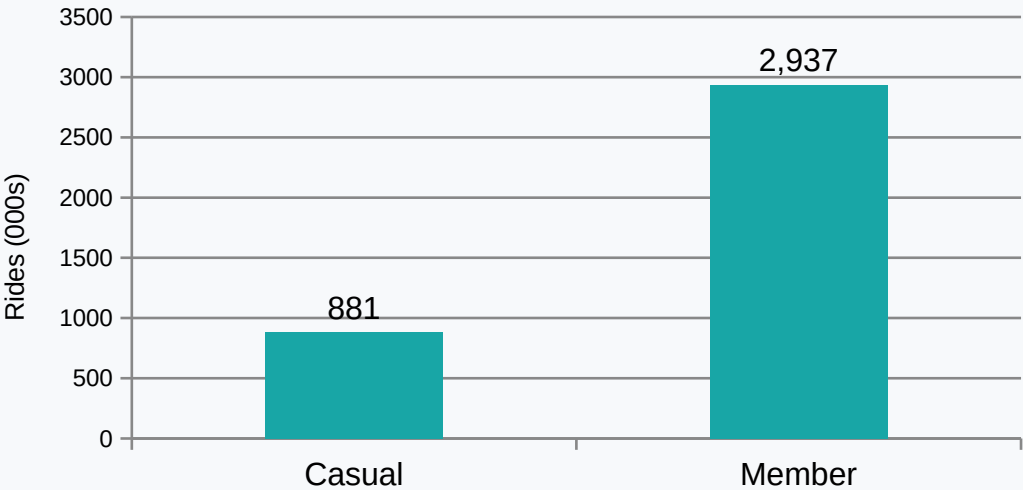
Case-study framing and required deliverables follow the provided Cyclistic course case study.

The two customer groups show distinct behavior

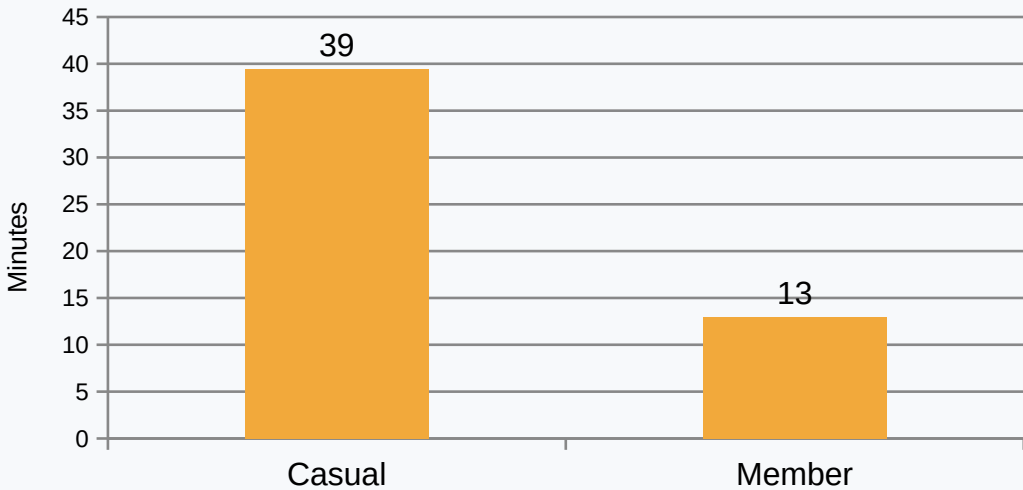
Members generate most trips; Casual riders take substantially longer rides.



Ride volume



Average ride length

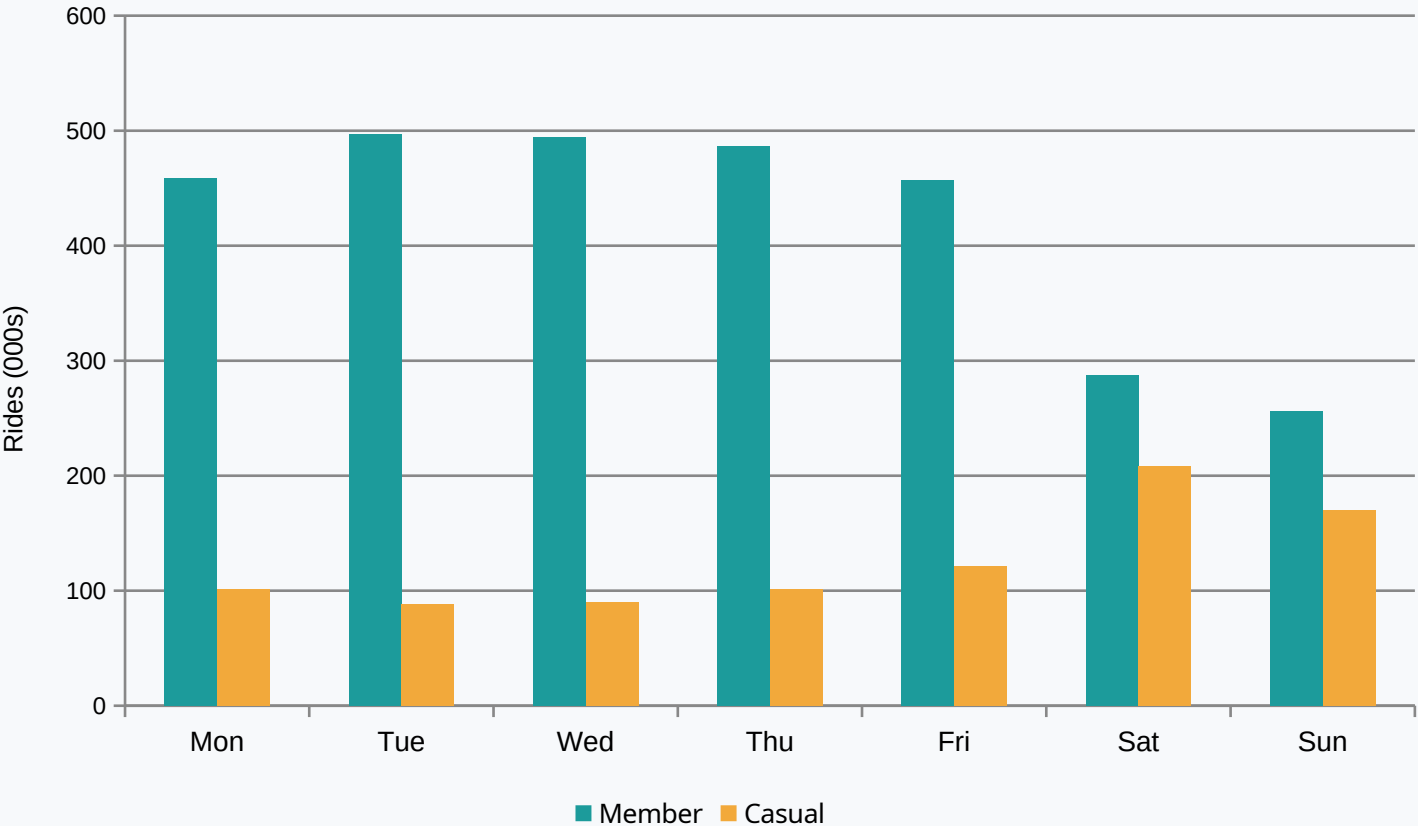


The case-study question is answered most clearly by comparing both frequency and ride duration.

Ride share uses all 3,818,004 trips. Average ride length uses 3,816,156 trips with valid duration (rides >24h excluded from that metric).

Weekly behavior separates the groups

Members are heavily weekday-oriented; Casual rides are more concentrated on weekends.



Average duration (valid rides) Casual: 37.90 → 41.45 min Member: 12.60 → 14.35 min

Ride counts use all trips by day. Weekend duration figures use records with valid ride length.

01

Thursday leads total weekly volume

588,287 rides — the highest daily total in 2019.

02

Casual weekend behavior is more prominent

Weekend Casual rides are much closer to weekday volume than they are for Members.

03

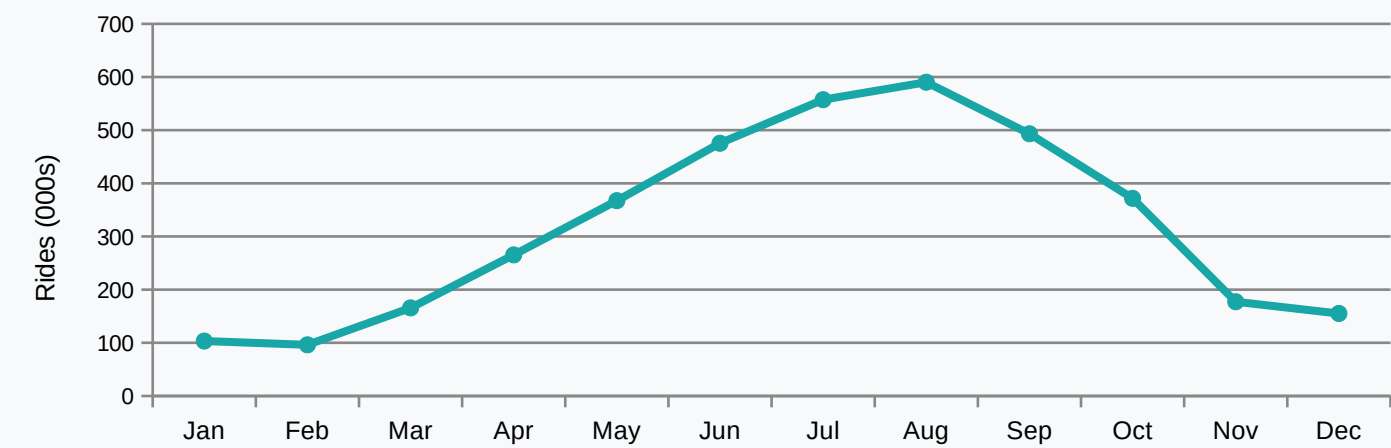
Ride duration rises on weekends

Average ride length increases for both groups on weekends.

Demand follows the calendar and the clock

Seasonality and an evening peak create clear windows for targeted messaging and capacity planning.

Monthly ride volume



01

August is the busiest month

590,184 rides — about 6× January volume.

02

July has the longest average ride

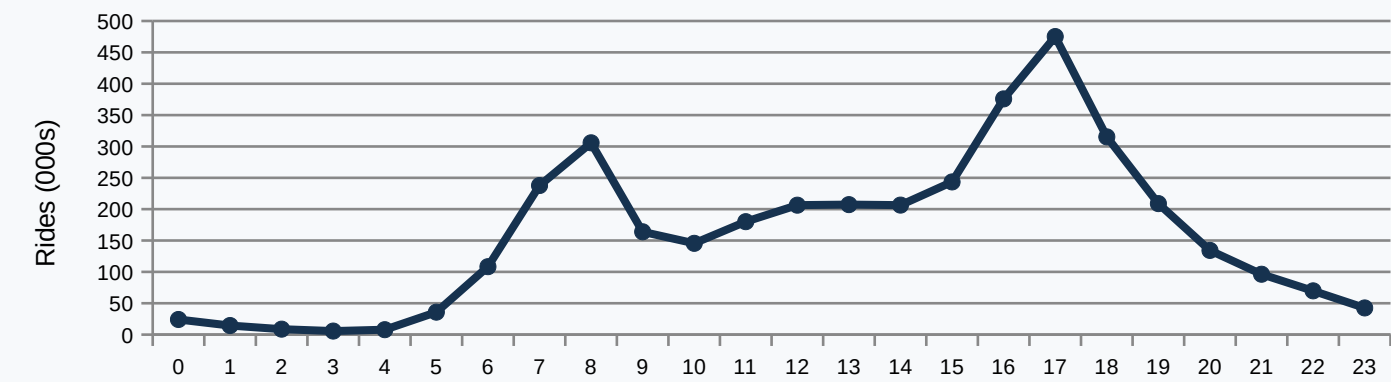
22.62 minutes, compared with 11.72 minutes in February.

03

5 PM is the peak hour

475,154 rides, the largest hourly total in the year.

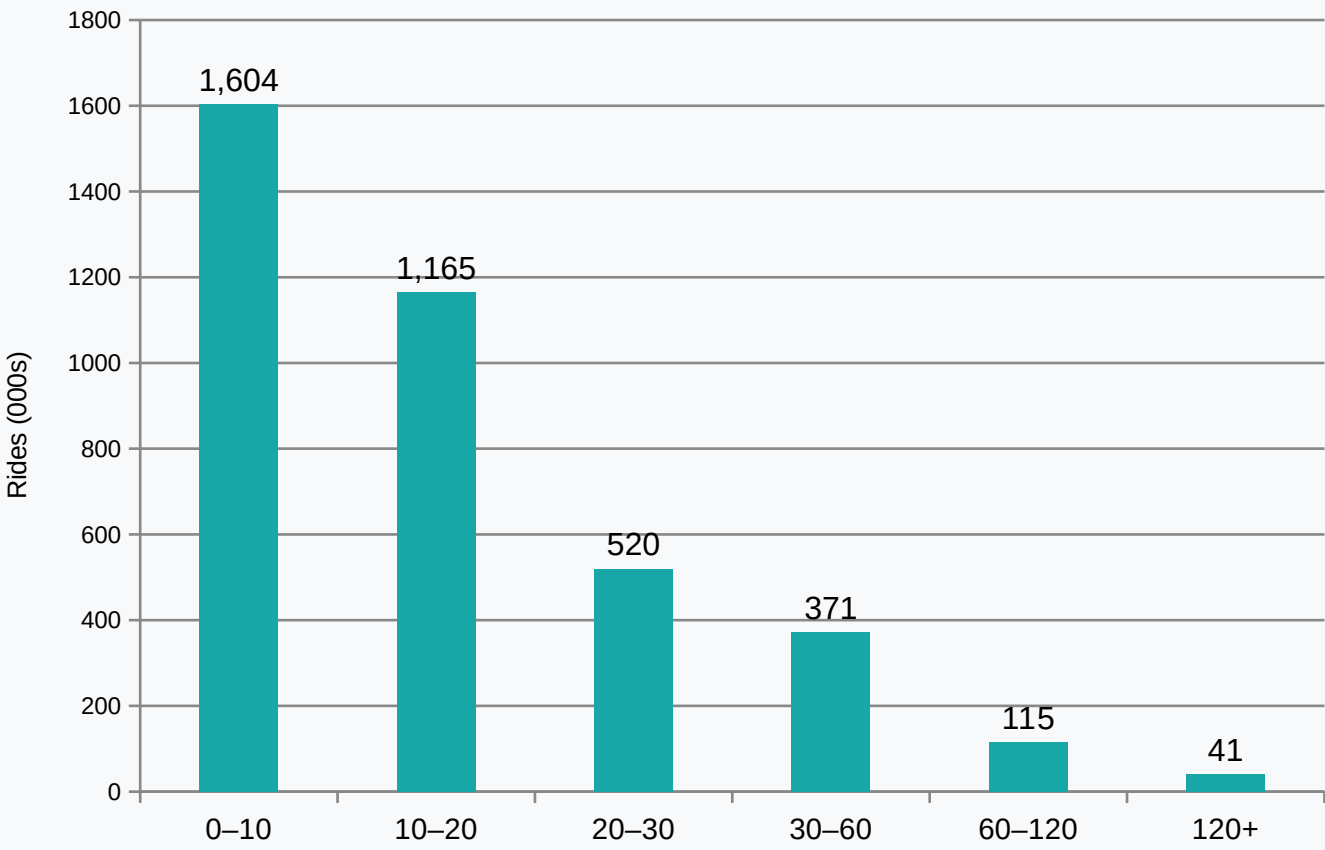
Hourly ride volume



Monthly and hourly totals use all 3,818,004 trips.

Most rides are short; Casual rides are consistently longer

Duration distribution gives context that a single average cannot.



72.6%

under 20 min

86.2%

under 30 min

01

Casual rides average 39.43 min
More than 3× the Member average of 12.93 min.

02

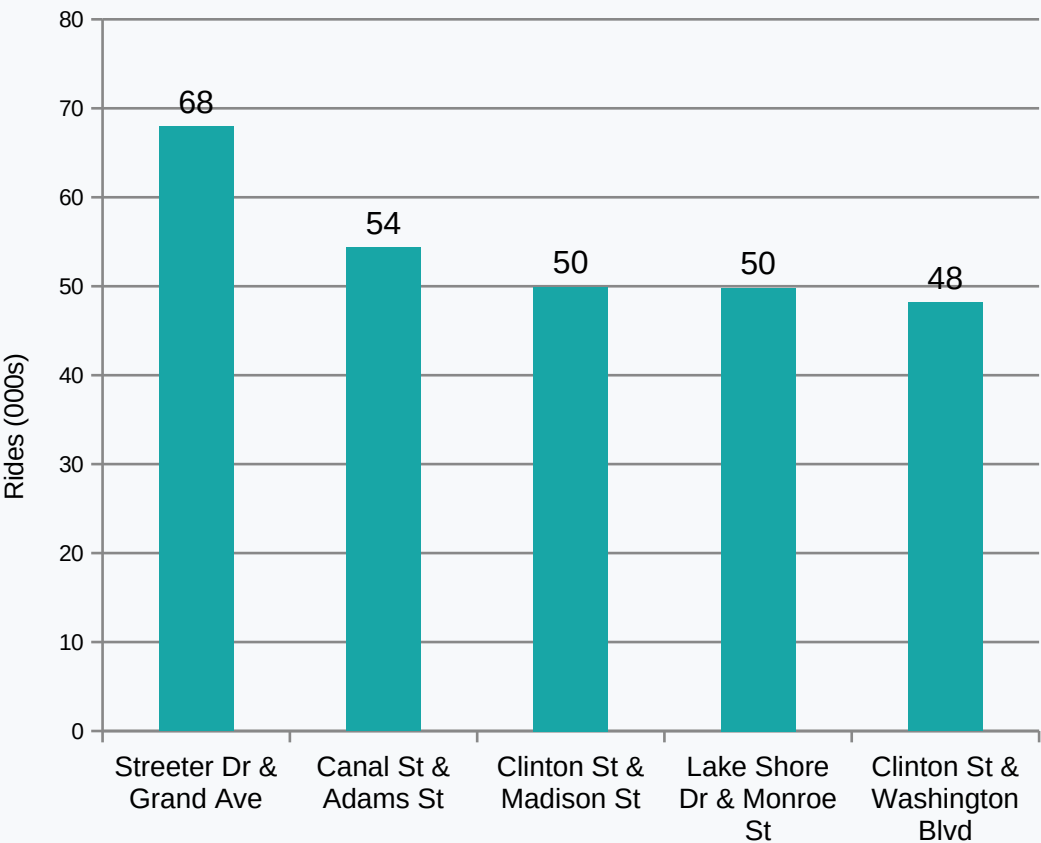
Weekend duration rises for both groups
Casual: 37.90 → 41.45 min. Member: 12.60 → 14.35 min.

Distribution covers 3,816,156 rides with valid duration; rides >24h are excluded from ride-length metrics.

Demand is concentrated at recognizable places and routes

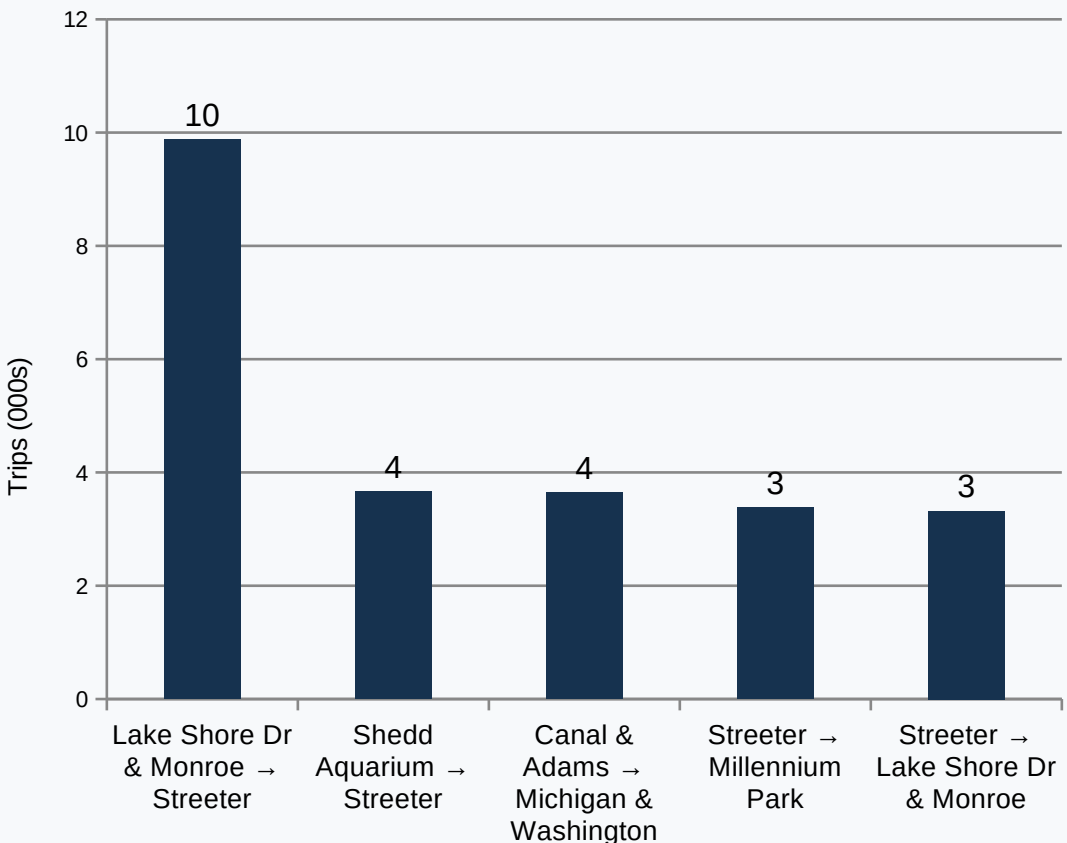
Station and route patterns can guide where to test conversion messaging.

Top starting stations



Streeter Dr & Grand Ave is #1 for both departures (67,983) and arrivals (81,723).

Top station-to-station routes



Top route: Lake Shore Dr & Monroe → Streeter (9,872 trips).

Station analysis excludes same-station trips from the Top Routes view.

From observed behavior to testable actions

Recommendations are framed as experiments because the trip data describes behavior, not conversion causes.

01

Focus conversion offers on Casual-heavy occasions

Test weekend and warm-season membership trials or targeted promotions, where Casual rides are more prominent and longer.

02

Test location-led messaging at high-demand destinations

Use high-traffic stations and routes—especially Streeter Dr & Grand Ave and adjacent routes—as test locations for membership messaging.

03

Time digital outreach around peak demand windows

Test late-afternoon/5 PM messaging and summer campaigns; measure click-to-membership conversion rather than assuming lift.

What should Cyclistic measure next?



Conversion rate

Casual → annual membership by campaign / location / timing



Offer response

Clicks, redemptions, trial starts, membership purchases



Economics

Cost per converted member and incremental lifetime value



Retention

90-day and 12-month retention of converted riders

Key limitation

No campaign, pricing, membership-conversion or rider-level purchase data was available, so these are hypotheses to test—not causal conclusions.